

MOHIT KUMAR

Security Engineer | AI for Security & Security for AI

mohit.kr1103@gmail.com | 9973395247 | Bengaluru, India | [LinkedIn](#) | [Github](#) | [X](#) | [Portfolio](#)

EDUCATION

Vellore Institute of Technology (VIT Chennai) Chennai, India
Bachelor of Technology in Electronics and Computer Engineering | **CGPA:** Aug 2022 – July 2026
9.12/10.0

Radiant International School Patna, India
Higher Secondary Education | **Class XII: 95.6%** | **Class X: 94.2%** 2019 – 2021

PROFESSIONAL EXPERIENCE

ITC Infotech India Ltd. Bengaluru, India (On-Site)
Security Engineer (IS2) 16 Jul 2026 – Present

- Promoted to full-time Security Engineer on completion of the internship, driving **AI-for-Security and Security-for-AI** initiatives — building autonomous detection tooling and hardening ML/agent supply chains alongside day-to-day SOC and VAPT operations.
- Supporting **GRC** engagements including **J-SOX** and **MICS** control auditing across IT general controls (access, change, and segregation-of-duties), testing control design and effectiveness and driving remediation of identified gaps to closure.

ITC Infotech India Ltd. Bengaluru, India (On-Site)
Cybersecurity Intern 05 Jan 2026 – 15 Jul 2026

- Designed and deployed a multi-agent AI email security system leveraging 7 parallel ML agents, achieving 96–97% threat detection accuracy with <0.8% miss rate and 1.3% false positives, significantly improving phishing detection reliability.
- Reduced SOC email triage time from 30 minutes to a few seconds by building a real-time, explainable decision pipeline with automated threat correlation, counterfactual analysis, and SOC-ready attack narratives.
- Engineered a low-latency (<100 ms), high-throughput distributed architecture with 99.9% system uptime, enabling scalable, cost-efficient phishing detection without GPU dependency while improving analyst productivity.

PROJECTS

Security Tooling Suite — 5 Open-Source Defensive Scanners | *Python, SARIF 2.1.0, CycloneDX, Offline & Deterministic*

- **Bulwark** — security stack for agentic AI (models, MCP servers, agent assemblies); catches **14/14** evasive pickle-RCE payloads, outperforming picklescan, modelscan and fickling, and emits a CycloneDX AI-BOM; 270+ tests.
- **Bastion** — Kubernetes RBAC attack-path analyzer; surfaced **47 privilege-escalation paths** to cluster-admin across 9 real Helm charts (368 manifests) where a popular linter found 0; 530+ tests.
- **Lattice** — post-quantum crypto-readiness scanner; builds a Cryptographic BOM across **12 languages**, grading quantum and classical risk against NIST FIPS 203/204/205 with zero runtime dependencies; 171 tests.
- **Portcullis** — CI/CD pipeline security scanner (GitHub Actions, GitLab CI, Jenkins) with trigger-context risk scoring; validated on 66 real pipelines and passes its own scan **100/100**; 130 tests.
- **Stowaway** — software supply-chain scanner (npm/PyPI/Go/Cargo) detecting typosquatting, dependency confusion and install-hook malware with **0 false positives** on 153 real dependencies; 133 tests.

Schema-Grounded NL→KQL for SIEM Rule Generation | *LLMs, LangGraph, ASIM, Microsoft Sentinel*

- Designed an intermediate-representation pipeline (schema validator + repair loop) that cut field-hallucination in LLM-generated Sentinel detection rules **from 93% to 13%** (~13× fewer, McNemar $p < 1e-7$), recovering 96.2% of failing cases within three repair attempts.

TECHNICAL SKILLS

Programming: Python, Java, SQL, Prolog, Bash

Security Engineering: AI/ML & Agent Security, Software Supply-Chain Security, Kubernetes & RBAC

Security, Post-Quantum Cryptography, DevSecOps & CI/CD Security, SIEM & Detection Engineering (Microsoft Sentinel/KQL, Splunk), VAPT (Burp Suite, Nmap, Nessus), SOC & Threat Hunting, Email Forensics (SPF/DKIM/DMARC), GRC (J-SOX, MICS), MITRE ATT&CK, SARIF, CycloneDX

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangGraph, LangChain, HuggingFace, XGBoost, OpenCV

Tools & Technologies: FastAPI, Docker, Git/GitHub, Azure AI, MongoDB, Streamlit, Jira, Jupyter

Relevant Coursework: Operating Systems, Computer Networks, DBMS, OOPs, Data Structures & Algorithms, Compiler Design, Deep Learning, Computer Vision, NLP

CERTIFICATIONS & ACHIEVEMENTS

Certifications: [IBM Cybersecurity Analyst Professional Certificate](#) | [Stanford Machine Learning Specialization](#) | [Microsoft Azure AI Engineer Associate](#) | [University of London Cyber Security Fundamentals](#) | [OCI Generative AI Professional](#) | [Azure AI Fundamentals](#)

Publications: **AgriCure: An AWS S3-Integrated Deep Learning Platform for Automated Crop Disease Detection** | *IEEE Xplore* | [Link to Paper](#) Dec 2025

Achievements: IEEE IC Hackathon 2.0 Finalist (AIR 7/600 teams) | 2nd Position State-level Science Exhibition | Coordinated Bihar's first Atal Tinkering Lab (Niti Aayog funded) | Languages: English, Hindi, Bengali